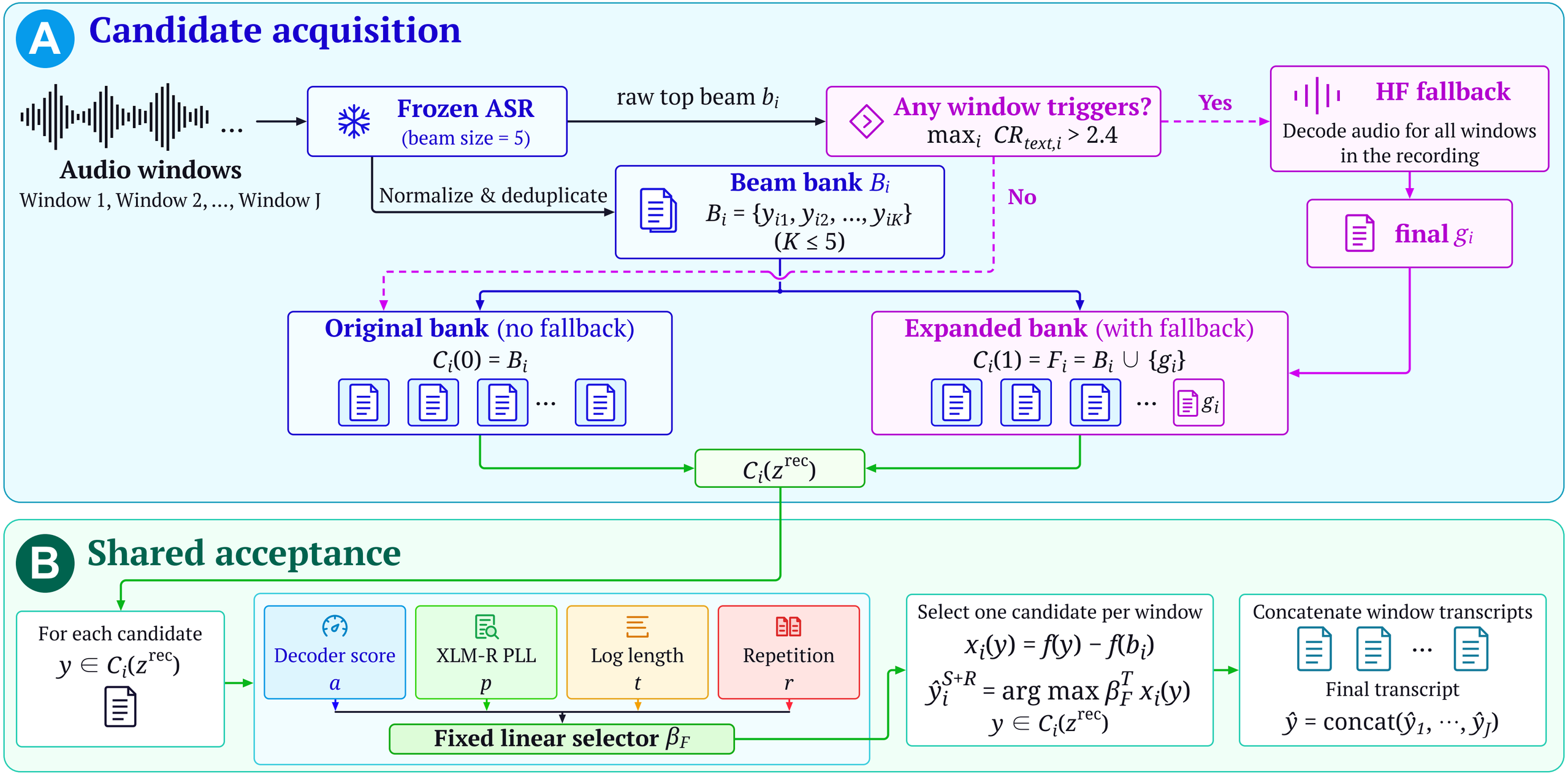

A
Candidate acquisition
Audio windows
Window 1, Window 2, ..., Window J
Frozen ASR
(beam size = 5)
raw top beam b_i
Any window triggers?
$\max_i CR_{text,i} > 2.4$
Yes
HF fallback
Decode audio for all windows in the recording
final g_i
No
Normalize & deduplicate
Beam bank B_i
$B_i = \{y_{i1}, y_{i2}, ..., y_{iK}\}$
$(K \le 5)$
Original bank (no fallback)
$C_i(0) = B_i$
Expanded bank (with fallback)
$C_i(1) = F_i = B_i \cup \{g_i\}$
$C_i(z^{rec})$
B
Shared acceptance
For each candidate
$y \in C_i(z^{rec})$
Decoder score
a
XLM-R PLL
p
Log length
t
Repetition
r
Fixed linear selector β_F
Select one candidate per window
$x_i(y) = f(y) - f(b_i)$
$\hat{y}_i^{S+R} = \arg\max \beta_F^T x_i(y)$
$y \in C_i(z^{rec})$
Concatenate window transcripts
Final transcript
$\hat{y} = \text{concat}(\hat{y}_1, \cdots, \hat{y}_J)$